Dihedral Groups, Klein Four-Group, Q8 and the Sylow Theorems

From the class book: Chapter 4, Section 4.2; Chapter 5, Sections 5.3 and 5.4; Chapter 6, Section 6.3; Chapter 13, Sections 13.1 to 13.4

1. Key definitions

Dihedral group. $D_n = \langle r, s \mid r^n = s^2 = e,\ srs^{-1} = r^{-1} \rangle = \{e, r, \ldots, r^{n-1}, s, sr, \ldots, sr^{n-1}\}$, the symmetries of a regular n-gon: order $2n$, with n rotations and n reflections. Non-abelian for $n \geq 3$; $D_3 \cong S_3$.

Klein four-group. $K_4 = \{e, a, b, c\}$ with $a^2 = b^2 = c^2 = e$, $ab = c,\ bc = a,\ ca = b$.
$K_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \cong U(8) \cong U(12)$; in S_4 it appears as $V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$.

Quaternion group. $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with $i^2 = j^2 = k^2 = -1$, $ij = k = -ji$, $jk = i = -kj$, $ki = j = -ik$. Presentation: $\langle a, b \mid a^4 = e,\ a^2 = b^2,\ b^{-1}ab = a^{-1} \rangle$. Matrix model: $i = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$, $j = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $k = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$.

Centre, centralizer, normalizer. $Z(G) = \{z : zx = xz\ \forall x\}$; $C(a) = \{x : xa = ax\}$; $N(H) = \{x : xHx^{-1} = H\}$. All are subgroups, and $|C(a)| = |G|/|\mathrm{cl}(a)|$.

p-group, Sylow subgroup. A p-group is a group in which every element has p-power order (finite: $|G| = p^k$). If $|G| = p^a m$, $p \nmid m$, a subgroup of order p^a is a Sylow p-subgroup; n_p denotes their number.

2. Main results

Theorem (Elements of D_n)
Every reflection sr^k has order 2; the rotation r^k has order $n/\gcd(k, n)$. Elements of order 2: n if n is odd, $n + 1$ if n is even (the extra one is $r^{n/2}$). For $k \mid n$, $k \geq 3$, the elements of order k are the $\varphi(k)$ rotations of order k. $Z(D_n) = \{e\}$ for n odd and $\{e, r^{n/2}\}$ for n even; $D_n' = \langle r^2 \rangle$.

Theorem (Subgroups of D_n)
D_n has $\tau(n) + \sigma(n)$ subgroups: $\tau(n)$ rotation subgroups $\langle r^d \rangle$ and $\sigma(n)$ subgroups containing reflections ($\langle r^d, sr^i \rangle$, $0 \leq i < d$, for each $d \mid n$). Of these, $\tau(n) + n$ are cyclic. Conjugacy classes: $(n + 3)/2$ for n odd, $(n + 6)/2$ for n even.

Theorem (Q_8)
One element of order 1, one of order 2 (-1), six of order 4. Subgroups: $\{1\}, \{\pm 1\}, \langle i \rangle, \langle j \rangle, \langle k \rangle, Q_8$ – 6, all normal, though Q_8 is non-abelian. $Z(Q_8) = Q_8' = \{\pm 1\}$, $Q_8/Z(Q_8) \cong K_4$, $\mathrm{Aut}(Q_8) \cong S_4$, $C(i) = \{\pm 1, \pm i\}$.

Theorem (Cauchy)
If a prime p divides $|G|$, then G has an element (hence a subgroup) of order p.

Theorem (Sylow)

Let $|G| = p^a m$, $p \nmid m$.

1. *Existence.* G has a subgroup of order p^k for every $p^k \mid |G|$; every p-subgroup lies in some Sylow p-subgroup.
2. *Conjugacy.* Any two Sylow p-subgroups are conjugate (hence isomorphic).
3. *Number.* $n_p \equiv 1 \pmod{p}$, $n_p \mid m$, and $n_p = [G : N(P)]$.

Consequently $P \trianglelefteq G \iff n_p = 1$; a finite abelian group is the direct product of its Sylow subgroups.

Theorem (Order pq, $p < q$ primes)

$n_q = 1$, so G is never simple. If $p \nmid q - 1$, $G \cong \mathbb{Z}_{pq}$ (orders 15, 33, 35, 51, 65, 77, 85, 95). If $p \mid q - 1$, there are exactly two groups: $\mathbb{Z}_{pq}$ and one non-abelian group (6: S_3; 21; $2q$: D_q; 39; 55). More generally, every group of order n is cyclic $\iff \gcd(n, \varphi(n)) = 1$.

Theorem (Index theorem)

If G has a proper subgroup H of index n with $|G| \nmid n!$, then G is not simple (the kernel of $G \to S_n$, acting on cosets of H, is a proper non-trivial normal subgroup inside H). A subgroup whose index is the smallest prime dividing $|G|$ is normal. A simple group with a subgroup of index $n > 1$ embeds in A_n; so in a non-abelian simple group every proper subgroup has index ≥ 5.

3. Standard tables

Number of subgroups

Group	Order	Subgroups	Remarks
$\mathbb{Z}_n$	n	$\tau(n)$	one per divisor
$\mathbb{Z}_p \times \mathbb{Z}_p$	p^2	$p + 3$	$p + 1$ of order p
K_4	4	5	three of order 2; all proper ones cyclic
$\mathbb{Z}_2 \times \mathbb{Z}_4$	8	8	
$\mathbb{Z}_2^3$	8	16	seven of order 2, seven of order 4
Q_8	8	6	all normal
$S_3 \cong D_3$	6	6	three of order 2, one of order 3
D_n	$2n$	$\tau(n) + \sigma(n)$	D_4: 10, D_5: 8, D_6: 16
A_4	12	10	$1 + 3 + 4 + 1 + 1$; none of order 6
S_4	24	30	$1 + 9 + 4 + 7 + 4 + 3 + 1 + 1$ (orders 1,2,3,4,6,8,12,24)

The 7 subgroups of order 4 in S_4: three cyclic $\langle (abcd) \rangle$, the normal V_4, and three Klein groups $\{e, (ab), (cd), (ab)(cd)\}$.

D_4 versus Q_8 (both non-abelian of order 8)

	elements of order 2	of order 4	subgroups	normal	centre	Aut
D_4	5	2	10	6	$\{e, r^2\}$	D_4 (order 8)
Q_8	1	6	6	6	$\{\pm 1\}$	S_4 (order 24)

Centres

Abelian G: G. S_n $(n \geq 3)$, A_n $(n \geq 4)$, D_n (n odd): $\{e\}$. D_n (n even): $\{e, r^{n/2}\}$. $GL(n, F)$: non-zero scalars. $SL(2, \mathbb{R})$: $\{\pm I\}$. $Z(G_1 \times G_2) = Z(G_1) \times Z(G_2)$.

Sylow numbers

Group (order)	n_2	n_3	n_5
S_3 (6)	3	1	–
A_4 (12)	1	4	–
S_4 (24)	3	4	–
A_5 (60)	5	10	6
S_5 (120)	15	10	6
D_n, n odd ($2n$)	n	–	–

If Sylow p-subgroups have order p, they meet trivially and give $n_p(p-1)$ elements of order p: e.g. A_5 has $6 \cdot 4 = 24$ elements of order 5 and $10 \cdot 2 = 20$ of order 3.

4. Exam shortcuts and standard counter-examples

Groups with all proper subgroups cyclic, yet not cyclic. K_4 (abelian), Q_8 and S_3 (non-abelian). K_4 is the smallest non-cyclic group. Q_8 is the standard non-abelian group with every subgroup normal.

Sylow routine. Write $|G| = p^a m$; list divisors of m that are $\equiv 1 \pmod{p}$. If some $n_p = 1$, G is not simple. Otherwise count elements of prime order: $\sum n_p(p-1) > |G|$ forces some $n_p = 1$. Then try the index theorem with $n = n_p$.

Never simple. Orders pq, p^2q, p^n $(n \geq 2)$, pqr (the Sylow r-subgroup, r largest, is normal), and $2m$ with $m > 1$ odd. No non-abelian simple group has order below 60; A_5 is the unique simple group of order 60; the next is $GL(3, \mathbb{Z}_2)$ of order 168 ($n_7 = 8$). In a group of order p^n, a subgroup of order p^{n-1} is normal.

Traps. The count $n_p(p-1)$ is valid only when Sylow subgroups have prime order. Sylow conditions give only possible values of n_p. Order pq is cyclic only if $p \nmid q - 1$ (S_3: $2 \mid 3 - 1$). Sylow subgroups need not be abelian ($D_4 \leq S_4$).

5. Solved examples

Example 1. For the dihedral group of order 20, find the number of elements of orders 2, 5, 10, the number of subgroups, and the order of the centre.

Solution. Here $n = 10$ (even). Order 2: 10 reflections $+ r^5 = 11$. Order 5: $\varphi(5) = 4$ rotations; order 10: $\varphi(10) = 4$ rotations. Check: $1 + 11 + 4 + 4 = 20$. Subgroups: $\tau(10) + \sigma(10) = 4 + 18 = 22$. $Z(D_{10}) = \{e, r^5\}$, of order 2.

Example 2. Show that a group of order 12 has a normal Sylow 2-subgroup or a normal Sylow 3-subgroup.

Solution. $12 = 4 \cdot 3$: $n_3 \mid 4, n_3 \equiv 1\ (3) \Rightarrow n_3 \in \{1, 4\}$; $n_2 \mid 3, n_2 \in \{1, 3\}$. If $n_3 = 4$, the four subgroups of order 3 give $4 \cdot 2 = 8$ elements of order 3. The remaining 4 elements must form the unique Sylow 2-subgroup, so $n_2 = 1$. Hence one of them is normal and no group of order 12 is simple. (A_4: $n_3 = 4, n_2 = 1$.)

Example 3. Prove that every group of order 105 has normal Sylow 5- and 7-subgroups.

Solution. $105 = 3 \cdot 5 \cdot 7$. $n_7 \mid 15$, $n_7 \equiv 1\ (7) \Rightarrow n_7 \in \{1, 15\}$; $n_5 \mid 21 \Rightarrow n_5 \in \{1, 21\}$; $n_3 \in \{1, 7\}$. If $n_7 = 15$ and $n_5 = 21$, there are $90 + 84 = 174 > 105$ elements – impossible. So P_5 or P_7 is normal, and $P_5 P_7$ is a subgroup of order 35 (cyclic, as $5 \nmid 6$). Its index 3 is the smallest prime divisor of 105, so it is normal. Its Sylow 5- and 7-subgroups are characteristic in it (unique of their orders), hence normal in G: $n_5 = n_7 = 1$.

Example 4. Is every group of order 77 cyclic? How many elements of order 77 does it have? What about order 39?

Solution. $77 = 7 \cdot 11$ and $7 \nmid 10$, so every group of order 77 is $\mathbb{Z}_{77}$ (equivalently $\gcd(77, \varphi(77)) = \gcd(77, 60) = 1$). Elements of order 77: $\varphi(77) = 6 \cdot 10 = 60$. For $39 = 3 \cdot 13$, $3 \mid 12$, so a non-abelian group of order 39 exists; not every group of order 39 is cyclic.

Example 5. Show that there is no simple group of order 200 or of order 36.

Solution. $200 = 8 \cdot 25$: $n_5 \mid 8$ and $n_5 \equiv 1\ (5)$; the divisors $1, 2, 4, 8$ leave only $n_5 = 1$, so the Sylow 5-subgroup is normal.
$36 = 4 \cdot 9$: $n_3 \in \{1, 4\}$. If $n_3 = 1$ we are done. If $n_3 = 4$, $N(P)$ has index 4, and $36 \nmid 4! = 24$; by the index theorem the conjugation action $G \to S_4$ has a proper non-trivial kernel. Either way G is not simple.

Example 6. Find n_2 and n_3 for S_4, identify a Sylow 2-subgroup, and count elements of order 3 in S_5 via Sylow theory.

Solution. $|S_4| = 8 \cdot 3$. Sylow 3-subgroups are $\langle (abc) \rangle$: 8 three-cycles give $8/2 = 4 = n_3$. Sylow 2-subgroups have order 8; $n_2 \in \{1, 3\}$, and $n_2 \neq 1$ since the 9 elements of order 2 and 6 of order 4 cannot fit in one subgroup of order 8. So $n_2 = 3$, each $\cong D_4$, e.g. $\langle (1234), (13) \rangle$. In S_5 ($120 = 3 \cdot 40$) Sylow 3-subgroups have order 3 and $n_3 = 10$, giving $10 \cdot 2 = 20$ elements of order 3 – matching the count of 3-cycles $\binom{5}{3} \cdot 2 = 20$.